

Computational Modeling and Simulation of Satellite Attitude Control Using Reaction Wheels

Mid-Evaluation Report

Problem Statement: Attitude Control of a Satellite Using Reaction Wheels
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Repository: <https://github.com/vivekswamy18092007/Reaction-wheel-attitude-control>

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All dynamics and control logic are implemented from first principles in Python (NumPy only).
No pre-built attitude-control toolboxes are used, in compliance with the problem-statement constraints.

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1 Introduction and Scope

A spacecraft in orbit has nothing external to push against, so any change in its orientation has to come from angular momentum exchanged internally. This is exactly what a reaction wheel does: by changing the spin rate of a motor-driven flywheel, an equal and opposite angular momentum is imparted to the spacecraft body, a direct consequence of Newton’s third law. This project presents a first-principles simulation of a CubeSat-scale Attitude Determination and Control System (ADCS) using such wheels, and demonstrates the satellite autonomously slewing from an arbitrary initial attitude to a commanded target attitude and holding it there.

The effort spans two tracks delivered together. The *CAD track* (Section 10) produces the 3U structure and the reaction-wheel assembly and, crucially, the mass properties the simulation depends on. The *simulation track* (Sections 2–12) derives and codes the dynamics, kinematics, allocation, and control law. In keeping with the problem statement, every physical and control equation is derived by hand and implemented from scratch in NumPy; no library that “directly solves” attitude control is used.

Reading guide. Sections 2–6 cover the reference mission, the architecture, and the full mathematical model, ending with a Lyapunov stability proof for the controller. Section 7 documents the integrator, Section 8 the verification methodology, and Section 9 the implementation status. Section 10 is the CAD track. Sections 11–12 record assumptions, limitations, and the roadmap to final submission.

2 Reference Mission and Physical Parameters

Before any dynamics we fix a concrete reference mission against which every performance target and disturbance estimate is measured.

Reference mission. A 3U CubeSat in a Sun-synchronous, near-circular low-Earth orbit performing nadir-pointing Earth observation. It exercises every requirement in the problem statement: a large-angle slew to acquire a target, followed by a precise hold against orbital disturbances.

Table 1: Reference orbit parameters ($R_{\oplus} = 6378.14 \text{ km}$, $\mu = 3.986 \times 10^5 \text{ km}^3/\text{s}^2$).

Quantity	Symbol	Value
Orbit altitude	h	500 km
Orbit radius	$r = R_{\oplus} + h$	6878.1 km
Orbital velocity	$v = \sqrt{\mu/r}$	7.613 km/s
Orbital period	$T = 2\pi\sqrt{r^3/\mu}$	5677 s $\approx$ 94.6 min
Mean motion	$n = 2\pi/T$	$1.107 \times 10^{-3} \text{ rad/s}$
Inclination (Sun-synchronous)	i	$\approx 97.4^\circ$

Mass properties. The inertia tensor feeding the dynamics is taken directly from the CAD assembly (Section 10) and expressed *about the centre of mass (CoM)* in the body frame. Because the assembly is very nearly four-fold symmetric, the CoM sits effectively at the geometric origin (offset $\sim 0.003 \text{ mm}$) and the body axes are already aligned with the principal axes, so the tensor is diagonal to within rounding. Only the CoM-referenced tensor is admissible in Euler’s equations (Section 4.3), and all wheel position vectors are referred to the CoM. The figures below are the final CAD-extracted values used directly by the dynamics model.

3 System Architecture and Data Flow

The simulation is built as a set of independent, separately testable modules, each mirroring one physical subsystem. Figure 1 traces a single simulation step: an outer loop advances time, while the inner block

Table 2: Mass properties and actuator parameters (CAD-extracted and design values).

Quantity	Symbol	Value
Total mass	m	1.765 kg
Transverse inertia	$I_{xx} = I_{yy}$	$1.401 \times 10^{-2} \text{ kg m}^2$, $1.413 \times 10^{-2} \text{ kg m}^2$
Spin-axis inertia	I_{zz}	$2.938 \times 10^{-3} \text{ kg m}^2$
Wheel inertia (each)	I_w	$1.1 \times 10^{-5} \text{ kg m}^2$
Wheel max speed	$\Omega_{\max}$	15,000 rpm = 1570.8 rad/s
Wheel max momentum	$h_{\max} = I_w \Omega_{\max}$	$1.728 \times 10^{-2} \text{ N m s}$
Wheel max torque	$\tau_{\max}$	$3.0 \times 10^{-2} \text{ N m}$

performs the fourth-order Runge–Kutta (RK4) update of the coupled kinematic and dynamic state. The commanded torque is held constant across the four RK4 sub-stages (zero-order hold), exactly as a real digital controller would drive the wheels between samples.

Module map: QuaternionLibrary (algebra, conversions) $\rightarrow$ QuaternionKinematics ($\dot{\mathbf{q}}$) and RotationalDynamics ($\dot{\boldsymbol{\omega}}$) $\rightarrow$ ReactionWheelArray (allocation, saturation, diagnostics) $\rightarrow$ PD Controller $\rightarrow$ AttitudeSimulator (RK4 driver, logger) $\rightarrow$ AttitudeVisualizer (3D animation, HUD). Each module can be unit-tested in isolation.

4 Mathematical Model

4.1 Attitude representation: unit quaternions

Euler angles are intuitive, but they have certain issues: at certain orientations two of the three axes align and a degree of freedom vanishes (gimbal lock), precisely during the large-angle slews this project must perform. We sidestep the problem entirely with a unit quaternion ,

$$\mathbf{q} = [q_0 \ q_1 \ q_2 \ q_3]^\top = [\cos(\theta/2) \ \hat{\mathbf{e}} \sin(\theta/2)]^\top, \quad \|\mathbf{q}\| = 1, \quad (1)$$

which encodes a rotation of angle θ about the eigenaxis $\hat{\mathbf{e}}$, is singularity-free, and needs only a cheap renormalisation to stay valid. The body-to-inertial rotation matrix is

$$\mathbf{R}(\mathbf{q}) = \begin{bmatrix} 1 - 2(q_2^2 + q_3^2) & 2(q_1q_2 - q_3q_0) & 2(q_1q_3 + q_2q_0) \\ 2(q_1q_2 + q_3q_0) & 1 - 2(q_1^2 + q_3^2) & 2(q_2q_3 - q_1q_0) \\ 2(q_1q_3 - q_2q_0) & 2(q_2q_3 + q_1q_0) & 1 - 2(q_1^2 + q_2^2) \end{bmatrix}. \quad (2)$$

The Hamilton product, conjugate, and inverse live in QuaternionLibrary; for a unit quaternion the inverse is simply the conjugate.

4.2 Attitude kinematics

Body angular velocity $\boldsymbol{\omega} = [\omega_x \ \omega_y \ \omega_z]^\top$ drives the quaternion through

$$\dot{\mathbf{q}} = \frac{1}{2} \Omega(\boldsymbol{\omega}) \mathbf{q}, \quad \Omega(\boldsymbol{\omega}) = \begin{bmatrix} 0 & -\omega_x & -\omega_y & -\omega_z \\ \omega_x & 0 & \omega_z & -\omega_y \\ \omega_y & -\omega_z & 0 & \omega_x \\ \omega_z & \omega_y & -\omega_x & 0 \end{bmatrix}. \quad (3)$$

The matrix $\Omega(\boldsymbol{\omega})$ is skew-symmetric, so $\frac{d}{dt} \|\mathbf{q}\|^2 = \mathbf{q}^\top \Omega(\boldsymbol{\omega}) \mathbf{q} = 0$ — the unit norm is conserved exactly in continuous time. Discretisation messes with this, which is why we renormalise each step (Section 7).

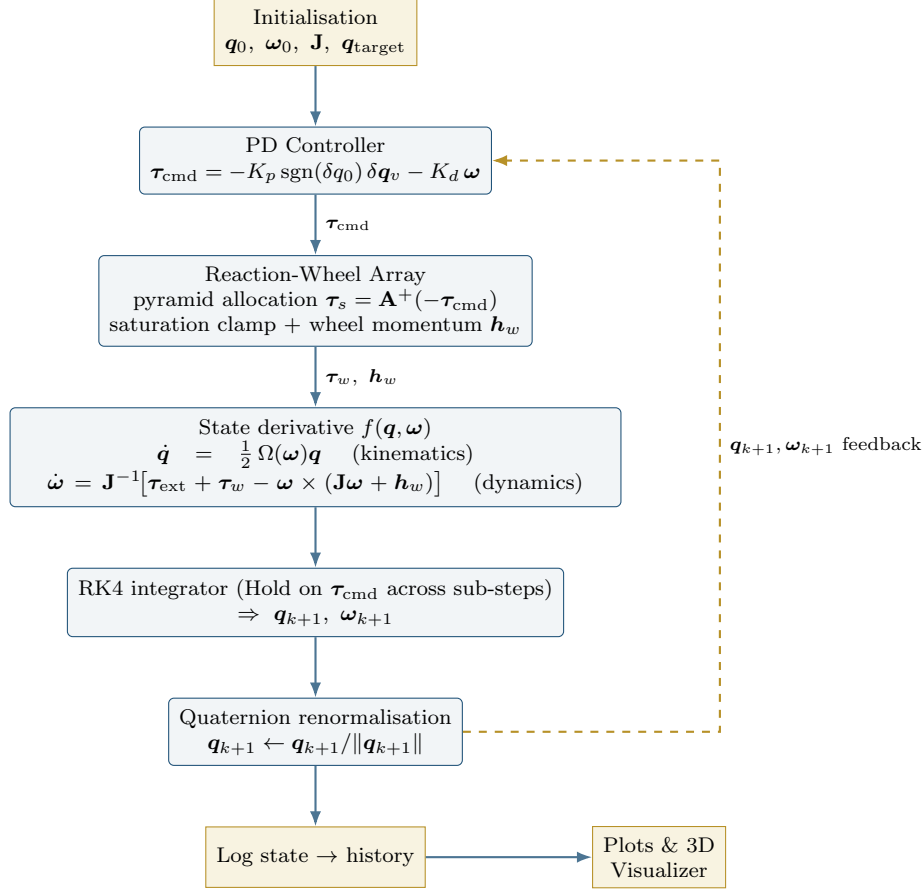


Figure 1: Closed-loop simulation data flow. Solid arrows: data path within one step. Dashed arrow: state feedback to the controller for the next step.

4.3 Rigid-body rotational dynamics with momentum exchange

The satellite is a rigid body rotating about its CoM. Because a free-flying body always rotates about its CoM and its rotation decouples from translation, only the CoM-referenced inertia tensor is physically admissible here.

Let $\mathbf{J}$ be the total inertia (bus plus wheels as part of the rigid body) and $\mathbf{h}_w \in \mathbb{R}^3$ the net wheel angular momentum in the body frame. Total system momentum about the CoM is $\mathbf{H} = \mathbf{J}\boldsymbol{\omega} + \mathbf{h}_w$. Newton's rotational law sets the inertial derivative of $\mathbf{H}$ equal to external torque; the transport theorem is used to rewrite it in the rotating body frame:

$$\left. \frac{d\mathbf{H}}{dt} \right|_{\text{inertial}} = \left. \frac{d\mathbf{H}}{dt} \right|_{\text{body}} + \boldsymbol{\omega} \times \mathbf{H} = \boldsymbol{\tau}_{\text{ext}}. \quad (4)$$

With $\dot{\mathbf{H}}|_{\text{body}} = \mathbf{J}\dot{\boldsymbol{\omega}} + \dot{\mathbf{h}}_w$ and the wheel reaction torque $\boldsymbol{\tau}_w \equiv -\dot{\mathbf{h}}_w$, the equation the dynamics module integrates is

$$\boxed{\mathbf{J}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau}_{\text{ext}} + \boldsymbol{\tau}_w - \boldsymbol{\omega} \times (\mathbf{J}\boldsymbol{\omega} + \mathbf{h}_w)} \quad (5)$$

The cross-product is the gyroscopic coupling: $\boldsymbol{\omega} \times \mathbf{J}\boldsymbol{\omega}$ is the classical Euler nutation term, and $\boldsymbol{\omega} \times \mathbf{h}_w$ is the extra coupling the spinning wheels inject. For the results in this report we omit the $\boldsymbol{\omega} \times \mathbf{h}_w$ contribution, a simplification that holds only while the stored wheel momentum $\mathbf{h}_w$ stays small compared with the body momentum $\mathbf{J}\boldsymbol{\omega}$, and stops being valid once the wheels spin up. Including this term in for the high-momentum, long-duration runs where it matters is part of our planned work (Section 12).

5 Reaction-Wheel Array: Geometry and Torque Allocation

While three orthogonal wheels provide minimum-redundancy 3-axis control, the loss of any single wheel eliminates control authority on one axis. A fourth wheel in a pyramid geometry makes the system rank-3 even with one wheel missing, preserving full attitude control despite single-wheel failure. This fault tolerance motivated our four-wheel design.

5.1 Four-wheel pyramid configuration

Four identical wheels sit in a pyramid: each spin axis is elevated by a fixed angle β above the body xy -plane and spaced 90° apart in azimuth ($\phi_i = 45^\circ, 135^\circ, 225^\circ, 315^\circ$), giving spin-axis unit vectors

$$\hat{\mathbf{e}}_i = [\cos \beta \cos \phi_i \quad \cos \beta \sin \phi_i \quad \sin \beta]^\top. \quad (6)$$

These form the columns of the distribution matrix $\mathbf{A} = [\hat{\mathbf{e}}_1 \ \hat{\mathbf{e}}_2 \ \hat{\mathbf{e}}_3 \ \hat{\mathbf{e}}_4] \in \mathbb{R}^{3 \times 4}$, and the momentum and torque the array delivers to the body are $\mathbf{h}_w = \mathbf{A}\mathbf{h}_s$ and $\boldsymbol{\tau}_w = -\mathbf{A}\boldsymbol{\tau}_s$, with $\mathbf{h}_s, \boldsymbol{\tau}_s \in \mathbb{R}^4$ the scalar per-wheel quantities.

Why the geometry is well-behaved. For the symmetric pyramid,

$$\mathbf{A}\mathbf{A}^\top = \text{diag}(2 \cos^2 \beta, 2 \cos^2 \beta, 4 \sin^2 \beta), \quad (7)$$

diagonal and well-conditioned thanks to the 90° azimuthal symmetry. Choosing $\beta \approx 35.26^\circ$ equalises authority across all three axes (it sets $2 \cos^2 \beta = 4 \sin^2 \beta$).

5.2 Minimum-norm torque allocation

Allocation runs the geometry backwards: given the controller's commanded body torque $\boldsymbol{\tau}_{\text{cmd}}$, find per-wheel torques with $-\mathbf{A}\boldsymbol{\tau}_s = \boldsymbol{\tau}_{\text{cmd}}$. Four unknowns, three equations and underdetermined by one. So we take the minimum-norm solution via the Moore–Penrose pseudo-inverse,

$$\boldsymbol{\tau}_s = \mathbf{A}^+(-\boldsymbol{\tau}_{\text{cmd}}), \quad \mathbf{A}^+ = \mathbf{A}^\top(\mathbf{A}\mathbf{A}^\top)^{-1}. \quad (8)$$

Minimising $\|\boldsymbol{\tau}_s\|$ shares the load across all four wheels rather than overworking one. This is provably identical to the Lagrange-multiplier solution of the constrained minimum-effort problem [2].

Saturation. Each wheel enforces $|h_{s,i}| \leq h_{\text{max}}$ and $|\tau_{s,i}| \leq \tau_{\text{max}}$; exceeding either clamps the command and logs the event alongside running diagnostics (per-wheel and total momentum, instantaneous power, accumulated energy). These feed both the saturation stress-test and the conservation checks.

6 Attitude Control Law

We are currently working on the control strategy for the CubeSat. Progress here has been slower than the rest of the project, mainly because of limited manpower on the team: with only two members splitting the CAD and simulation tracks, control-law development has had to wait behind getting the dynamics, kinematics, and reaction-wheel modules working and verified first. What follows is the basic quaternion-feedback PD law we have implemented as a starting point, along with the Lyapunov argument that justifies its structure. We have not yet tuned or validated this controller in closed-loop simulation. We are committing to a complete, properly tuned, and validated control strategy – including measured performance against the problem statement's scenarios – by the final evaluation.

6.1 From PID to PD

The proposal named a PID controller; on reflection, the integral term is unnecessary for the core task. For rigid-body *regulation* (slew to a fixed target and hold) there is no constant bias to integrate away, and a well-tuned proportional–derivative law already drives the error to zero with no steady-state offset. Integral action mainly fights persistent disturbance torques and brings its own wind-up headaches, so we adopt a quaternion-feedback PD law as the baseline and defer integral augmentation to the disturbance-rejection extension.

6.2 Error quaternion and control law

With current attitude $\mathbf{q}$ and target $\mathbf{q}_t$, the error quaternion is

$$\delta\mathbf{q} = \mathbf{q}_t^{-1} \otimes \mathbf{q} = [\delta q_0 \quad \delta \mathbf{q}_v^\top]^\top, \quad (9)$$

with $\delta q_0 \rightarrow 1$, $\delta \mathbf{q}_v \rightarrow \mathbf{0}$ at alignment. The commanded torque is

$$\boxed{\boldsymbol{\tau}_{\text{cmd}} = -K_p \text{sgn}(\delta q_0) \delta \mathbf{q}_v - K_d \boldsymbol{\omega}} \quad K_p, K_d > 0. \quad (10)$$

The vector part $\delta \mathbf{q}_v$ provides proportional restoring torque; $\boldsymbol{\omega}$ provides rate damping. The $\text{sgn}(\delta q_0)$ factor handles the quaternion double-cover ($\mathbf{q}$ and $-\mathbf{q}$ are the same orientation) so the controller always takes the short way round, never “unwinding” through a needless near-360° slew.

6.3 Stability justification

The structure is not arbitrary, but it admits a short Lyapunov argument. With the energy-like candidate

$$V = \frac{1}{2} \boldsymbol{\omega}^\top \mathbf{J} \boldsymbol{\omega} + 2K_p(1 - \delta q_0) \geq 0, \quad (11)$$

which is zero only at the target, differentiating along the closed-loop dynamics and substituting the control law (10) causes the proportional cross-terms to cancel exactly, leaving

$$\dot{V} = -K_d \|\boldsymbol{\omega}\|^2 \leq 0. \quad (12)$$

Since $\dot{V}$ vanishes only at $\boldsymbol{\omega} = \mathbf{0}$, LaSalle’s invariance principle gives asymptotic stability. We treat this as justification for the *form* of the law rather than a finished analysis: a closed-loop simulation with tuned gains, together with the second-order ω_n/ζ relationship used to select them, is the next step and is covered under Future Work (Section 12).

7 Numerical Integration

Euler integration just takes one slope estimate per step and runs with it, so errors pile up fast. Not great for something as accuracy over time. RK4 checks the slope at four points within each step and blends them, so it stays accurate without needing a small step size. accuracy effect, we still– something something

Even though for a slow, smooth slew the accuracy difference probably wouldn’t show up much in practice, RK4 was still the obvious choice. It costs almost nothing extra to implement and gives us a much bigger safety margin if the dynamics ever get faster or more aggressive. The coupled state $\mathbf{x} = (\mathbf{q}, \boldsymbol{\omega})$ evolves under $\dot{\mathbf{x}} = f(\mathbf{x}, \boldsymbol{\tau})$ from (3) and (5), advanced by fixed-step RK4: $\mathbf{x}_{k+1} = \mathbf{x}_k + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4)$, with global error $\mathcal{O}(\Delta t^4)$. Two disciplines matter:

- **Zero-order hold.** The commanded torque is frozen across all four RK4 sub-stages and recomputed from feedback only at the next step, faithfully modelling a once-per-sample digital controller rather than an unphysical continuously-varying one.
- **Renormalisation.** Each step ends with $\mathbf{q} \leftarrow \mathbf{q}/\|\mathbf{q}\|$, erasing the tiny unit-norm drift of finite-step integration and keeping $\mathbf{R}(\mathbf{q})$ a proper rotation. The residual $|\|\mathbf{q}\| - 1|$ doubles as a verification metric.

8 Model Verification Methodology

To check the reliability of the model we verify it. The following checks run in the torque-free, wheel-free configuration (the cleanest test of the bare propagator) unless noted.

1. **Angular-momentum conservation.** With no external torque and no wheels, the inertial momentum $\mathbf{H}_{\text{inertial}} = \mathbf{R}(\mathbf{q}) \mathbf{J} \boldsymbol{\omega}$ must be constant; we track its drift. Expected: bounded near machine precision.
2. **Energy conservation.** Kinetic energy $T = \frac{1}{2} \boldsymbol{\omega}^\top \mathbf{J} \boldsymbol{\omega}$ is conserved torque-free; we track $|T(t) - T(0)|/T(0)$.
3. **Norm drift.** $||\mathbf{q}|| - 1$ should stay within acceptable limits with renormalization on.
4. **Known motion.** A spin about a principal axis stays a pure spin; a torque-free asymmetric body shows the expected intermediate-axis instability and bounded polhode.
5. **Step convergence.** Halving Δt should shrink residuals at RK4's fourth-order rate.
6. **Controlled sanity.** With the controller on, V decreases monotonically and total system momentum is conserved as it transfers between body and wheels.

```
Run length: 200 s, dt = 0.01 s, w0 = [0.05, 0.3, 0.02] rad/s
Max |dH| over run : 1.251e-16 N.m.s
Max energy drift : 1.028e-14 (fractional)
Max |norm(q)-1| : 2.220e-16
Convergence dts : [0.2 0.1 0.05 0.025 0.0125 0.00625]
Convergence errors : [1.42681921e-11 8.89153658e-13 5.54935592e-14 3.46805486e-15
2.14613247e-16 1.82198393e-17]
```

9 Software Implementation Status

The codebase is public, organised into `src/` and `docs/`. Table 3 summarises module status at the mid-evaluation.

Table 3: Module implementation status.

Module	Status	Notes
QuaternionLibrary	Complete	Algebra, Euler $\leftrightarrow$ quaternion, rotation matrix.
QuaternionKinematics	Complete	$\dot{\mathbf{q}} = \frac{1}{2} \Omega(\boldsymbol{\omega}) \mathbf{q}$.
RotationalDynamics	Complete	Euler's equation; full $\mathbf{h}_w$ gyroscopic coupling being folded in for long-duration controlled runs.
RK4.Propagator	Complete	Coupled RK4 with ZOH and per-step renormalisation; documented in LaTeX.
AttitudeSimulator	Complete	4-wheel pyramid (primary) and 3-wheel orthogonal; pseudo-inverse allocation, saturation, diagnostics; documented in LaTeX.
ReactionWheelArray	Complete	
PD Controller	In progress	Quaternion-feedback PD law implemented and Lyapunov-justified (Section 6); closed-loop tuning and scenario testing not yet performed.
AttitudeVisualizer	Complete	3D model, body/inertial axes, ghost target, attitude-error HUD; documentation in progress.

10 CAD Design and Mechanical Configuration (CAD Track)

This section documents the mechanical design that defines the physical satellite and supplies the mass properties the simulation depends on.

10.1 Structural configuration and form factor

The selection of the 3U CubeSat form factor was driven primarily by two considerations: the accommodation of the pyramidal reaction wheel assembly and the desire to maintain industry relevance and realism. Preliminary studies indicated that smaller form factors impose severe constraints on the packaging of a four-wheel pyramidal configuration, so the 3U form factor ultimately won out. Furthermore, the increased external surface area available in a 3U configuration provides significant advantages for power generation from solar panels.

The chassis was designed in accordance with the Cal Poly CubeSat Design Specification, adopting the standard external dimensions of $100\text{ mm} \times 100\text{ mm} \times 340.5\text{ mm}$. The primary load-bearing members consist of four longitudinal rails having cross-sectional dimensions of $8.5\text{ mm} \times 8.5\text{ mm}$, as prescribed by the specification. Following Detail A of the reference blueprint, chamfers were introduced along the rail edges to ensure compliance with deployer interface requirements. Fillets were also applied to the external rail corners to reduce stress concentrations and improve manufacturability.

Trusses are used on the sides of the form factor because of their lightweight and structurally sound design. Solid panels provide excellent stiffness but result in unnecessary mass penalties. Circular cut-outs offer moderate mass reduction; however, they do not distribute loads as efficiently. Honeycomb structures provide superior weight savings but are considerably more complex to manufacture and model within the scope of this project. The body-axis convention aligns with the simulation track: z runs along the long axis of the 3U stack.

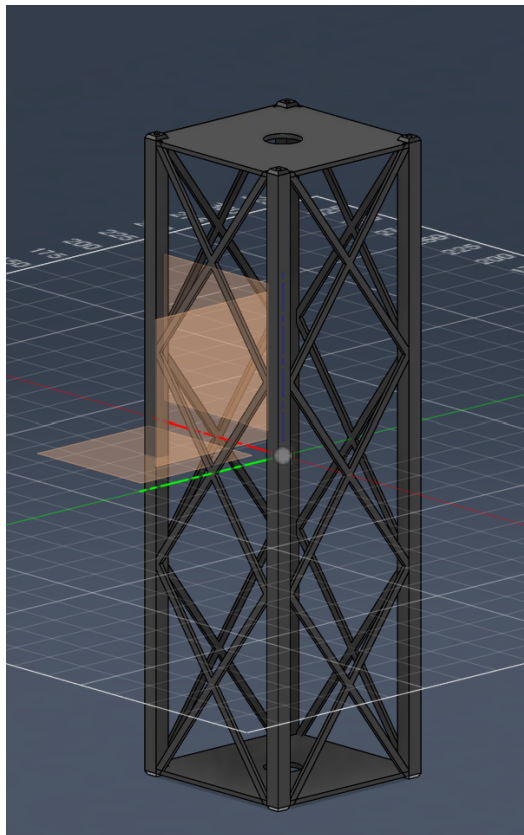


Figure 2: Isometric render of the 3U CubeSat chassis with body axes shown (red: x , green: y , blue: z). Truss side panels and longitudinal rail structure are visible.

10.2 Reaction-wheel assembly and mounting

The actuator selected for each reaction wheel is the **Maxon ECX Flat 22 L brushless DC motor**, chosen for its compact flat form factor and high torque output, both essential for fitting four units inside the 3U envelope without violating the CubeSat rail clearances. The flywheel itself is 38 mm in diameter and 10 mm thick, with a stainless-steel rim. A hub-spoke-rim topology is adopted: concentrating mass at the rim maximises the polar moment of inertia about the spin axis for a given total mass, yielding a wheel geometry whose spin-axis inertia I_w is close to the theoretical optimum for a thin annulus.

Each wheel is mounted at an elevation angle of $\beta = 32^\circ$ above the body xy -plane, with the four units spaced 90° apart in azimuth ($\phi_i = 45^\circ, 135^\circ, 225^\circ, 315^\circ$), reproducing the spin-axis unit vectors $\hat{e}_i$ of Section 5 exactly as designed. The as-built angle matches the nominal $\beta = 32^\circ$ with no deviation; note that this differs slightly from the equal-authority ideal of $\beta \approx 35.26^\circ$, a deliberate trade that provides marginally greater z -axis authority at the cost of a small asymmetry in the $\mathbf{AA}^\top$ diagonal.

The bracket uses gussets and an angled seating plane to hold each motor housing at the correct elevation. The motor housing in turn retains the Maxon motor via flanges that clamp against the bracket face, ensuring a rigid, repeatable interface. This two-piece bracket-plus-housing arrangement allows each unit to be removed and replaced independently without disturbing the remaining three wheels.

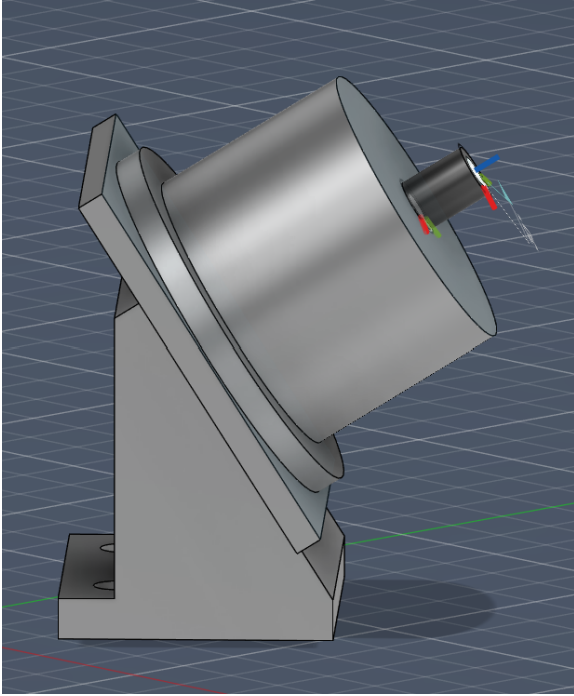


Figure 3: Side view of a single reaction-wheel bracket and motor-housing assembly. The angled seating plane sets $\beta = 32^\circ$; gussets stiffen the bracket against reaction torques. Flanges on the motor housing secure the Maxon ECX Flat 22 L to the bracket.

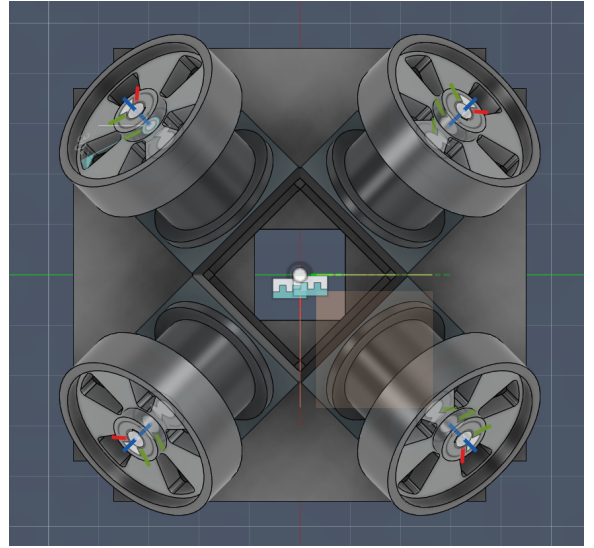


Figure 4: Top view of the full four-wheel pyramid assembly. The 90° azimuthal spacing ($\phi_i = 45^\circ, 135^\circ, 225^\circ, 315^\circ$) and the central mounting plate are clearly visible.

10.3 Material selection and mass budget

Materials were selected to balance structural performance, mass efficiency, and manufacturability. Aluminium alloys dominate the load-bearing structure: 6061 is used where weldability and general machinability are prioritised, while the higher-strength 7075 alloy is used in regions subject to greater stress concentrations. The flywheel rim is stainless steel to maximise rim density and therefore spin-axis inertia. The avionics and payload board stack is modelled as FR4 PCB with assorted components. The resulting total mass of 1.74 kg

is well within the realistic 3U range of 1–6 kg.

Table 4: Component mass budget (from CAD).

Component	Material	Mass
Structural frame	Aluminium 6061 & 7075	478 g
Flywheel ($\times 4$)	Aluminium 7075 (hub/spokes), stainless steel (rim)	26.2 g
Reaction wheel assembly	Aluminium 6061 & 7075	306.75 g
Avionics / payload (dummy)	FR4 & assorted materials	954.4 g
Total		1765.35 g

10.4 Inertia tensor extraction and centre of mass

This is the key handoff to the simulation track. The inertia tensor was exported from Fusion 360 in units of g mm^2 and converted to kg m^2 (multiply by 10^{-9}). The centre of mass lies at (0.002, 0.002, ≈ 0) mm relative to the geometric origin, an offset of 0.003 mm, which is negligible and consistent with the near-perfect four-fold symmetry of the assembly. The off-diagonal products of inertia are five or more orders of magnitude smaller than the smallest diagonal term ($|I_{xy}|/I_{\min} \approx 0.005\%$), confirming that the body frame is already aligned with the principal axes to within 0.076° , well below any modelling threshold.

Table 5: CAD-extracted inertia tensor about the CoM.

Quantity	Symbol	Value
Inertia tensor (CoM)	$\mathbf{J}_{\text{CoM}}$	$\begin{bmatrix} 1.4010 \times 10^{-2} & 1.598 \times 10^{-7} & 3.2 \times 10^{-11} \\ 1.598 \times 10^{-7} & 1.4130 \times 10^{-2} & 3.1 \times 10^{-11} \\ 3.2 \times 10^{-11} & 3.1 \times 10^{-11} & 2.9380 \times 10^{-3} \end{bmatrix} \text{ kg m}^2$
CoM offset from origin	$\mathbf{r}_{\text{CoM}}$	(0.002, 0.002, ≈ 0) mm — negligible
Principal-axis misalignment	—	0.076° (effectively zero)

10.5 Exploded view and assembly drawings

The assembly follows a bottom-up sequence. The structural frame: four longitudinal aluminium rails joined by truss side panels and three deck panels (bottom, middle, and top) forms the skeleton into which all internal components slot. The payload telescope mounts through the bottom deck and occupies the lower bay. The four reaction-wheel assemblies, each consisting of an inclined mounting bracket ($\beta = 32^\circ$), a Maxon ECX Flat 22 L motor, and a hub-spoke-rim flywheel, are then fitted onto the central mounting plate and lowered into the middle bay as a pre-assembled unit. The OBC board stacks above the wheel assembly on the middle deck, followed by the top deck panel which closes the structure.

Figure 5 shows the complete integrated assembly; the reaction-wheel pyramid is visible through the transparent truss panels alongside the avionics stack and payload bay. Figure 6 is the labelled exploded view showing every major component and its location within the stack.

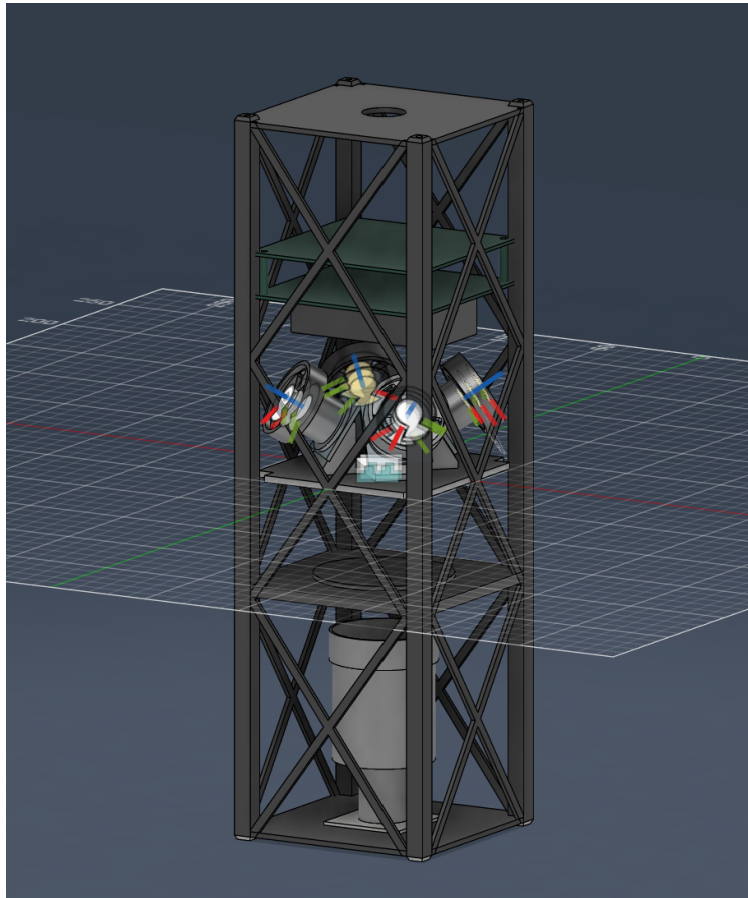


Figure 5: Full 3U CubeSat assembly render. The four-wheel pyramid occupies the central bay; the payload telescope protrudes from the bottom deck and the OBC board stack sits above the wheel assembly.

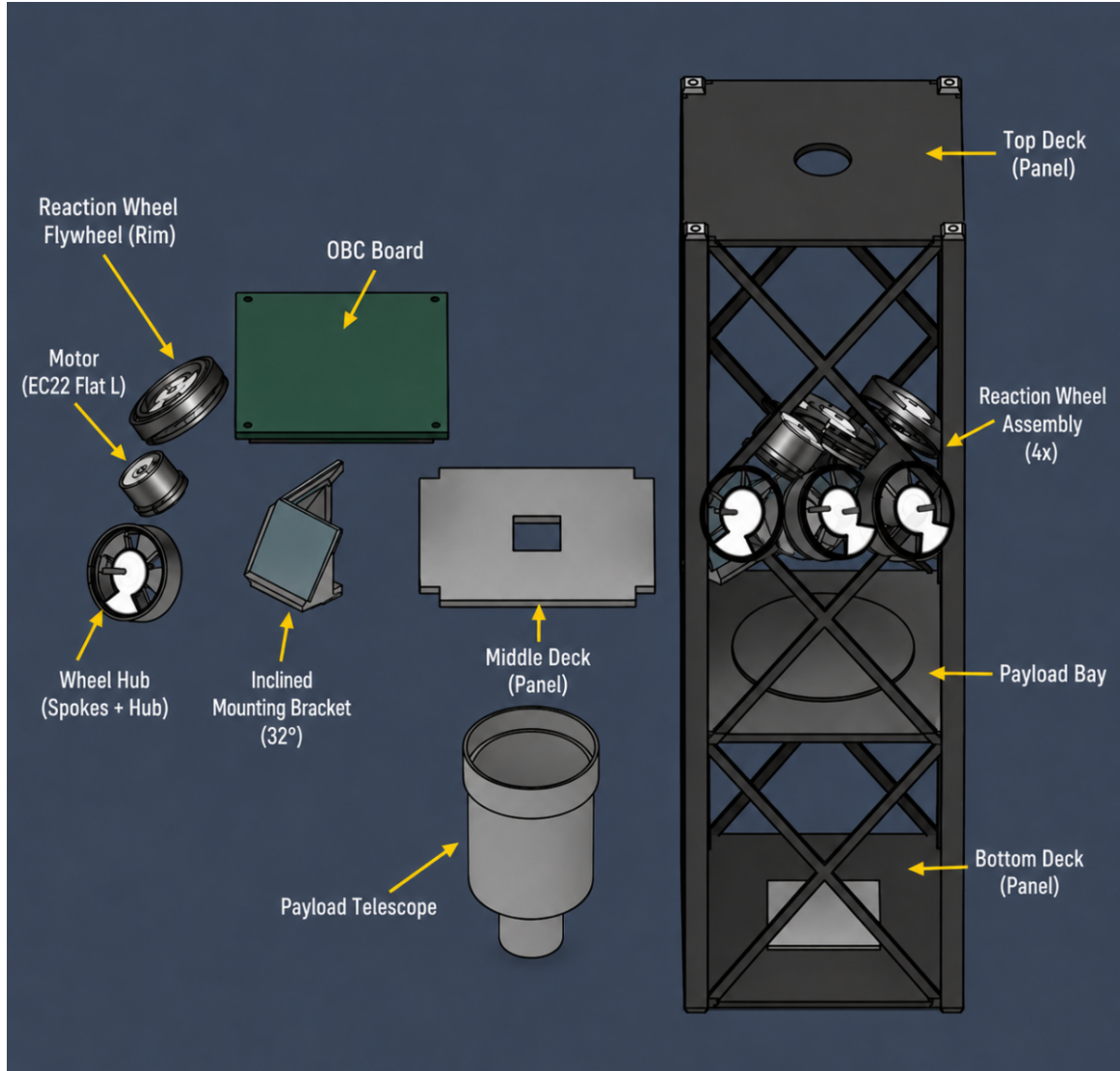


Figure 6: Labelled exploded view of the 3U assembly. Key components: reaction-wheel flywheel (rim + hub/spokes), Maxon ECX Flat 22L motor, inclined mounting bracket (32°), middle and top deck panels, OBC board, payload telescope, and the integrated assembly showing the reaction-wheel array ($\times 4$) and payload bay.

11 Assumptions and Limitations

- Single rigid body; structural flexibility, slosh, and micro-vibration are neglected.
- Wheels are ideal momentum-exchange actuators with hard speed/torque limits; motor torque-speed curves, bearing friction, and torque quantisation are deferred.
- Attitude determination (sensor fusion / estimation) is out of scope per the problem statement; the controller is given true state.
- Translation-rotation decoupling about the CoM, valid for a free-flying body whose actuators exert no net force.
- The pure-PD law leaves a small steady-state offset under persistent disturbance, which is a known and deliberate limitation, resolved by the planned integral augmentation.

- A reaction-wheel system cannot shed net momentum on its own: under a one-sided disturbance the wheels saturate and a secondary actuator (e.g. magnetorquers) is needed to desaturate. We model this boundary but do not implement desaturation hardware.

12 Future Work and Roadmap

Prioritised work toward final submission:

1. Tune the PD gains via the ω_n/ζ relationship in Section 6 and run the full closed-loop simulation against the problem statement’s two scenarios (large-angle slew and hold; disturbance rejection), reporting settling time, overshoot, steady-state error, and wheel momentum margin.
2. Fold the full $\mathbf{h}_w$ gyroscopic-coupling term into the dynamics for long controlled runs; complete the quantitative verification suite.
3. One-wheel-failure study: re-allocate across the surviving three wheels via the pseudo-inverse and quantify retained authority.
4. Eigenaxis vs. sequential-axis slew comparison; saturation stress-test.
5. Gravity-gradient disturbance model (peak $\tau_{gg} \sim \frac{3}{2}n^2|I_{\max} - I_{\min}| \approx 6 \times 10^{-8} \text{ N m}$ at the reference orbit) included in the disturbance-rejection scenario.
6. Second controller (LQR) for a comparative study; sliding-mode control as a stretch goal.

Longer-horizon: integral augmentation (PID) for bias rejection, a Monte-Carlo robustness study, and an interactive tuning dashboard.

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